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LEARNING FROM OBSERVATIONS

In which we describe agents that can improve their behavior through diligent study of their own experiences.

The idea behind learning is that percepts should be used not only for acting, but also for improving the agent's ability to act in the future. Learning takes place as a result of the interaction between the agent and the world, and from observation by the agent of its own decision-making processes. Learning can range from trivial memorization of experience, as exhibited by the wumpus agent in Chapter 7, to the creation of entire scientific theories, as exhibited by Albert Einstein. This chapter starts with the design of general learning agents, and describes inductive learning—constructing a description of a function from a set of input/output examples. We then give several algorithms for inductive learning in logical agents and a theoretical analysis that explains why learning works.

18.1 A GENERAL MODEL OF LEARNING AGENTS

LEARNING ELEMENT
PERFORMANCE
ELEMENT

A learning agent can be divided into four conceptual components, as shown in Figure 18.1. The most important distinction is between the **learning element**, which is responsible for making improvements, and the **performance element**, which is responsible for selecting external actions. The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions. The learning element takes some knowledge about the learning element and some feedback on how the agent is doing, and determines how the performance element should be modified to (hopefully) do better in the future. The design of the learning element depends very much on the design of the performance element. When trying to design an agent that learns a certain capability, the first question is not "How am I going to get it to learn this?" but "What kind of performance element will my agent need to do this once it has learned how?" For example, the learning algorithms for producing rules for logical systems are quite different from the learning algorithms for producing belief networks. We will see, however, that the *principles* behind the learning algorithms are much the same.

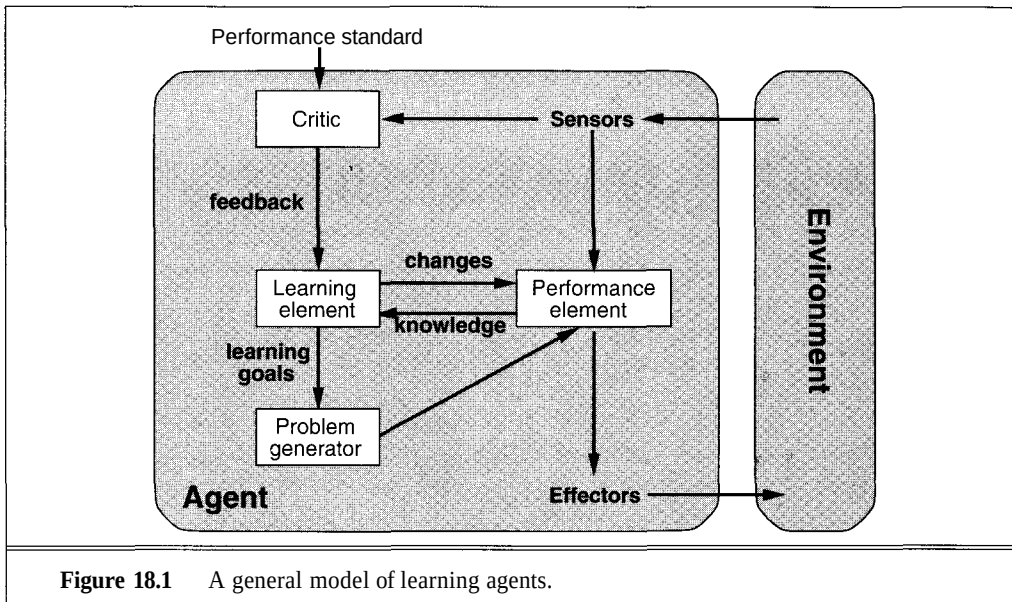


Figure 18.1 A general model of learning agents.

CRITIC

The **critic** is designed to tell the learning element how well the agent is doing. The critic employs a fixed standard of performance. This is necessary because the percepts themselves provide no indication of the agent's success. For example, a chess program may receive a percept indicating that it has checkmated its opponent, but it needs a performance standard to know that this is a good thing; the percept itself does not say so. It is important that the performance standard is a fixed measure that is conceptually outside the agent; otherwise the agent could adjust its performance standards to meet its behavior. In humans, this form of irrationality is called "sour grapes" and is characterized by comments such as "Oh well, never mind, I didn't want that stupid Nobel prize anyway."

PROBLEM
GENERATOR

The last component of the learning agent is the **problem generator**. It is responsible for suggesting actions that will lead to new and informative experiences. The point is that if the performance element had its way, it would keep doing the actions that are best, given what it knows. But if the agent is willing to explore a little, and do some perhaps suboptimal actions in the short run, it might discover much better actions for the long run. The problem generator's job is to suggest these exploratory actions. This is what scientists do when they carry out experiments. Galileo did not think that dropping rocks from the top of a tower in Pisa was valuable in itself; he was not trying to break the rocks, nor to render pedestrians unconscious. His aim was to demonstrate a better theory of the motion of objects.

To make the overall design more concrete, let us return to the automated taxi example. The performance element consists of whatever collection of knowledge and procedures the taxi has for selecting its driving actions (turning, accelerating, braking, honking, and so on). The taxi goes out on the road and drives, using this performance element. The learning element formulates goals, for example, to learn better rules describing the effects of braking and accelerating, to learn the geography of the area, to learn how the taxi behaves on wet roads, and to learn what

causes annoyance to other drivers. The critic observes the world and passes information along to the learning element. For example, after the taxi makes a quick left turn across three lanes of traffic, the critic observes the shocking language used by other drivers, the learning element is able to formulate a rule saying this was a bad action, and the performance element is modified by installing the new rule. Occasionally, the problem generator kicks in with a suggestion: try taking 7th Avenue uptown this time, and see if it is faster than the normal route.

The learning element is also responsible for improving the *efficiency* of the performance element. For example, when asked to make a trip to a new destination, the taxi might take a while to consult its map and plan the best route. But the next time a similar trip is requested, the planning process should be much faster. This is called **speedup learning**, and is dealt with in Chapter 21. In this chapter, we concentrate on the acquisition of knowledge.

SPEEDUP LEARNING

Machine learning researchers have come up with a large variety of learning elements. To understand them, it will help to see how their design is affected by the context in which they will operate. The design of the learning element is affected by four major issues:

- Which *components* of the performance element are to be improved.
- What *representation* is used for those components.
- What *feedback* is available.
- What *prior information* is available.

Components of the performance element

We have seen that there are many ways to build the performance element of an agent. The components can include the following:

1. A direct mapping from conditions on the current state to actions.
2. A means to infer relevant properties of the world from the percept sequence.
3. Information about the way the world evolves.
4. Information about the results of possible actions the agent can take.
5. Utility information indicating the desirability of world states.
6. Action-value information indicating the desirability of particular actions in particular states.
7. Goals that describe classes of states whose achievement maximizes the agent's utility.

Each of the components can be learned, given the appropriate feedback. For example, if the agent does an action and then perceives the resulting state of the environment, this information can be used to learn a description of the results of actions (4). If the taxi exerts a certain braking pressure when driving on a wet road, then it will soon find out how much actual deceleration is achieved. Similarly, if the critic can use the performance standard to deduce utility values from the percepts, then the agent can learn a useful representation of its utility function (5). If the taxi receives no tips from passengers who have been thoroughly shaken up during the trip, it can learn a useful component of its overall utility function. In a sense, the performance standard can be seen as defining a set of *distinguished percepts* that will be interpreted as providing direct feedback on the quality of the agent's behavior. Hardwired performance standards such as pain and hunger in animals can be understood in this way.

Representation of the components

Any of these components can be represented using any of the representation schemes in this book. We have seen several examples: deterministic descriptions such as linear weighted polynomials for utility functions in game-playing programs and propositional and first-order logical sentences for all of the components in a logical agent; and probabilistic descriptions such as belief networks for the inferential components of a decision-theoretic agent. Effective learning algorithms have been devised for all of these. The details of the learning algorithm will be different for each representation, but the main idea remains the same.

Available feedback

For some components, such as the component for predicting the outcome of an action, the available feedback generally tells the agent what the correct outcome is. That is, the agent predicts that a certain action (braking) will have a certain outcome (stopping in 10 feet), and the environment immediately provides a percept that describes the actual correct outcome (stopping in 15 feet). Any situation in which both the inputs and outputs of a component can be perceived is called **supervised learning**. (Often, the outputs are provided by a friendly teacher.)

On the other hand, in learning the condition-action component, the agent receives some evaluation of its action (such as a hefty bill for rear-ending the car in front) but is not told the correct action (to brake more gently and much earlier). This is called **reinforcement learning**; the hefty bill is called a **reinforcement**.¹ The subject is covered in Chapter 20.²

Learning when there is no hint at all about the correct outputs is called **unsupervised learning**. An unsupervised learner can always learn relationships among its percepts using supervised learning methods—that is, it can learn to predict its future percepts given its previous percepts. It cannot learn what to *do* unless it already has a utility function.

Prior knowledge

The majority of learning research in AI, computer science, and psychology has studied the case in which the agent begins with no knowledge at all about what it is trying to learn. It only has access to the examples presented by its experience. Although this is an important special case, it is by no means the general case. Most human learning takes place in the context of a good deal of background knowledge. Some psychologists and linguists claim that even newborn babies exhibit knowledge of the world. Whatever the truth of this claim, there is no doubt that prior knowledge can help enormously in learning. A physicist examining a stack of bubble-chamber photographs may be able to induce a theory positing the existence of a new particle of a certain mass and charge; but an art critic examining the same stack might learn nothing more than that the "artist" must be some sort of abstract expressionist. In Chapter 21, we see several ways in which learning is helped by the use of existing knowledge.

¹ The terms **reward and punishment** are also used as synonyms for **reinforcement**.

² Drawing the line between supervised and reinforcement learning is somewhat arbitrary; reinforcement learning can also be thought of as supervised learning with a less informative feedback signal.

SUPERVISED
LEARNING

REINFORCEMENT
LEARNING
REINFORCEMENT

UNSUPERVISED
LEARNING

Bringing it all together

Each of the seven components of the performance element can be described mathematically as a **function**: for example, information about the way the world evolves can be described as a function from a world state (the current state) to a world state (the next state or states); a goal can be described as a function from a state to a Boolean value (0 or 1) indicating whether the state satisfies the goal. The key point is that *all learning can be seen as learning the representation of a function*. We can choose which component of the performance element to improve and how it is to be represented. The available feedback may be more or less useful, and we may or may not have any prior knowledge. The underlying problem remains the same.

